

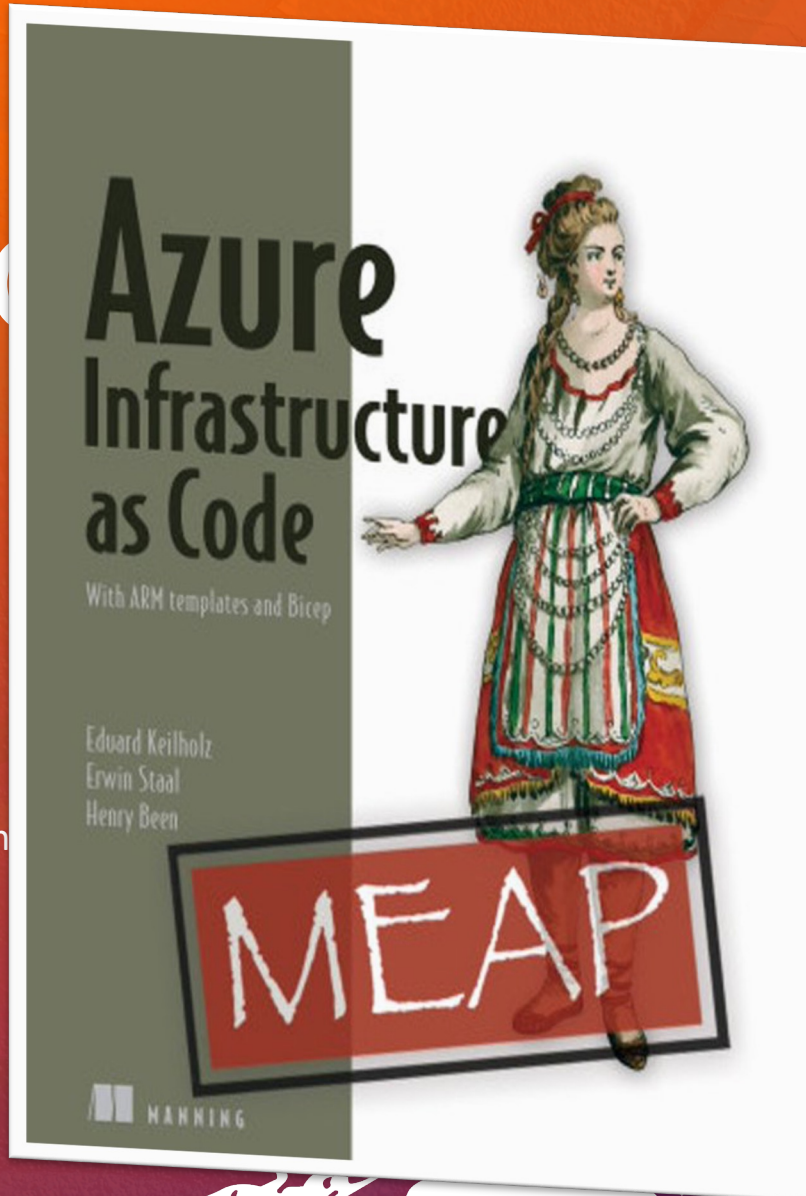
Infrastructure as Code on Azure: Bicep vs Terraform vs Pulumi



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Xebia | Xpirit



Meet the contestants!



Simplicity



Community



Ecosystem



Deployment



Developer experience



“Bicep is a domain-specific language (DSL) that uses declarative syntax to deploy Azure resources.”

- › Microsoft, 2021
- › Replacement for ARM templates
- › Open Source
- › The replacement of ARM Templates
- › DSL



“Terraform is a tool for building, changing, and versioning infrastructure safely and efficiently. Terraform can manage existing and popular service providers as well as custom in-house solutions.”

- › HashiCorp, 2014
- › ~~Open Source~~, Business Source License
- › DSL (HCL = HashiCorp Configuration Language)
- › * Terraform CDK (Cloud Development Kit)



Pulumi

“Pulumi provides the industry’s only automation workflow capability that allows software engineering to be applied to solve and manage cloud infrastructure at scale.”

- › Pulumi, 2018
- › Open Source
- › Write you IaC in your favorite language: TypeScript, JavaScript, Python, Go, C#, F#, Visual Basic, Java, and YAML



Demo!

Deploy with a plan

#Infra Core Transformation Spoke Pipeline • Merged PR 9258

Cancel

Summary

Environments

Terraform Plan

Associated pipelines

WhiteSource Bolt Build Report

Triggered by

View 15 changes

Repository and version

Time started and elapsed

Related

Tests and coverage

main 22e40841

Fri at 15:59
5d 18h 24m

4 work items
4 published

Get started

2 approvals need your review before this run can continue

Review

Stages

Jobs

Build

1 job completed 16s

1 artifact

Plan: pndt - ddev

1 job completed 6m 39s

1 artifact

Apply: pndt - ddev

1 job completed 49s

Plan: pndt - dprd

1 job completed 5m 59s

1 artifact

Apply: pndt - dprd

Waiting

0/1 checks passed

Plan: pndt - uprd

1 job completed 6m 49s

1 artifact

Apply: pndt - uprd

Waiting

0/1 checks passed

Deploy with a plan

#Infra Core Storage Spoke Pipeline • Merged PR 11927:

ⓘ This run will be cleaned up after 1 month based on your project settings.

Summary Environments **Terraform Plan** Associated pipelines WhiteSource Bolt Build Report

```
- site_config {
  - application_insights_connection_string = (sensitive value) -> null
  - application_insights_key              = (sensitive value) -> null
  # (23 unchanged attributes hidden)

  # (5 unchanged blocks hidden)
}
# (1 unchanged block hidden)
}

# module.compute.module.function_app.azurelinux_function_app.function_app will be updated in-place
- resource "azurerm_linux_function_app" "function_app" {
  - app_settings = {
    + "APPINSIGHTS_INSTRUMENTATIONKEY" = (sensitive value)
    + "APPLICATIONINSIGHTS_CONNECTION_STRING" = (sensitive value)
    # (8 unchanged elements hidden)
  }
  id = "/subscriptions/87db002d-529e-42d4-98a6-29ab47ea9c30/resourceGroups/rg-
  name = "func-we-pnds-comp-ddev-001"
  tags = {
    "CostCenter" =
    "environment" = "ddev"
  }
  # (25 unchanged attributes hidden)

  - site_config {
    - application_insights_connection_string = (sensitive value) -> null
    - application_insights_key              = (sensitive value) -> null
    # (23 unchanged attributes hidden)

    # (5 unchanged blocks hidden)
  }
  # (1 unchanged block hidden)
}

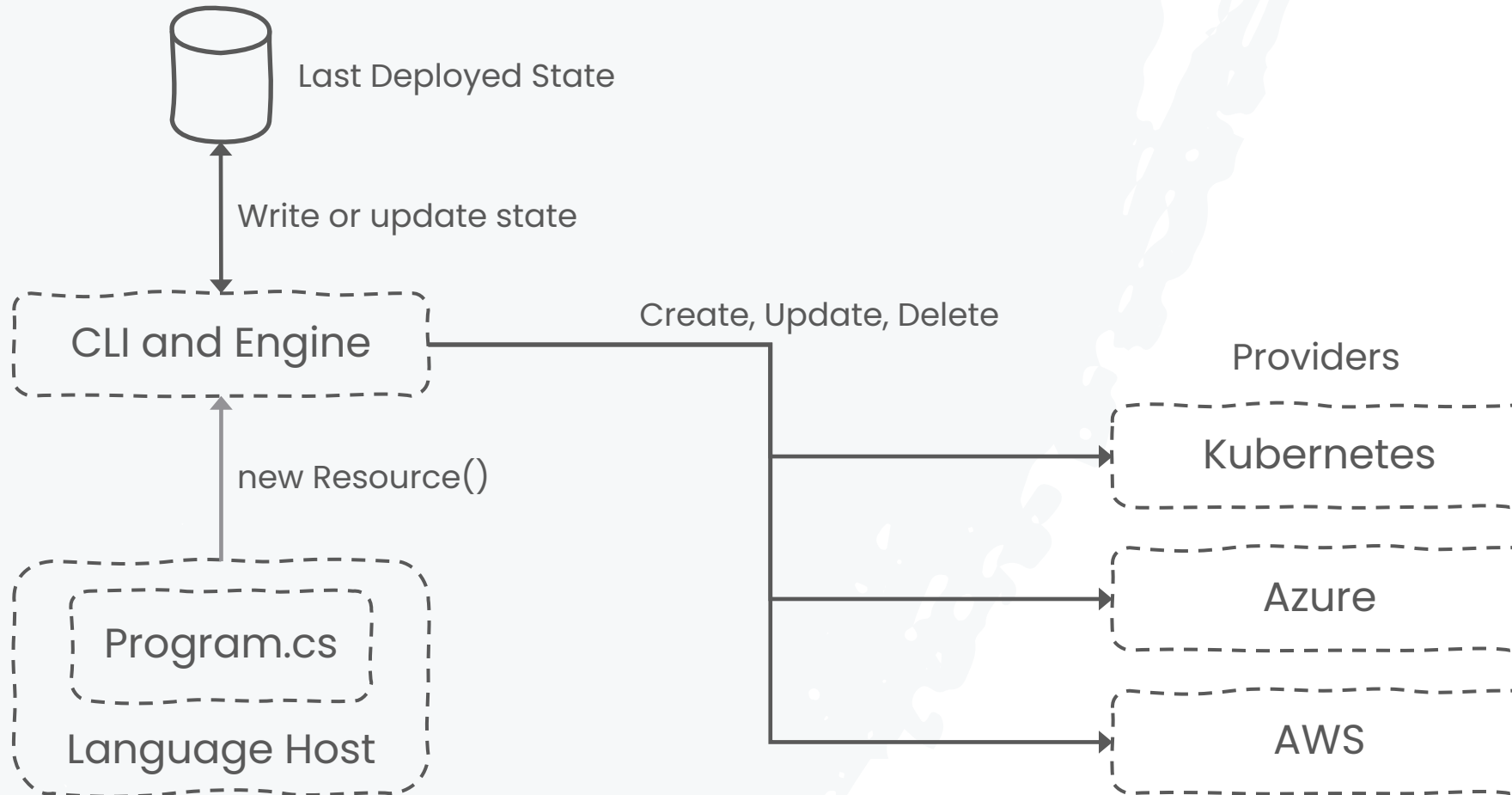
Plan: 0 to add, 3 to change, 0 to destroy.
```

Saved the plan to: pnds-ddev-104056-20231008001054-tf.plan

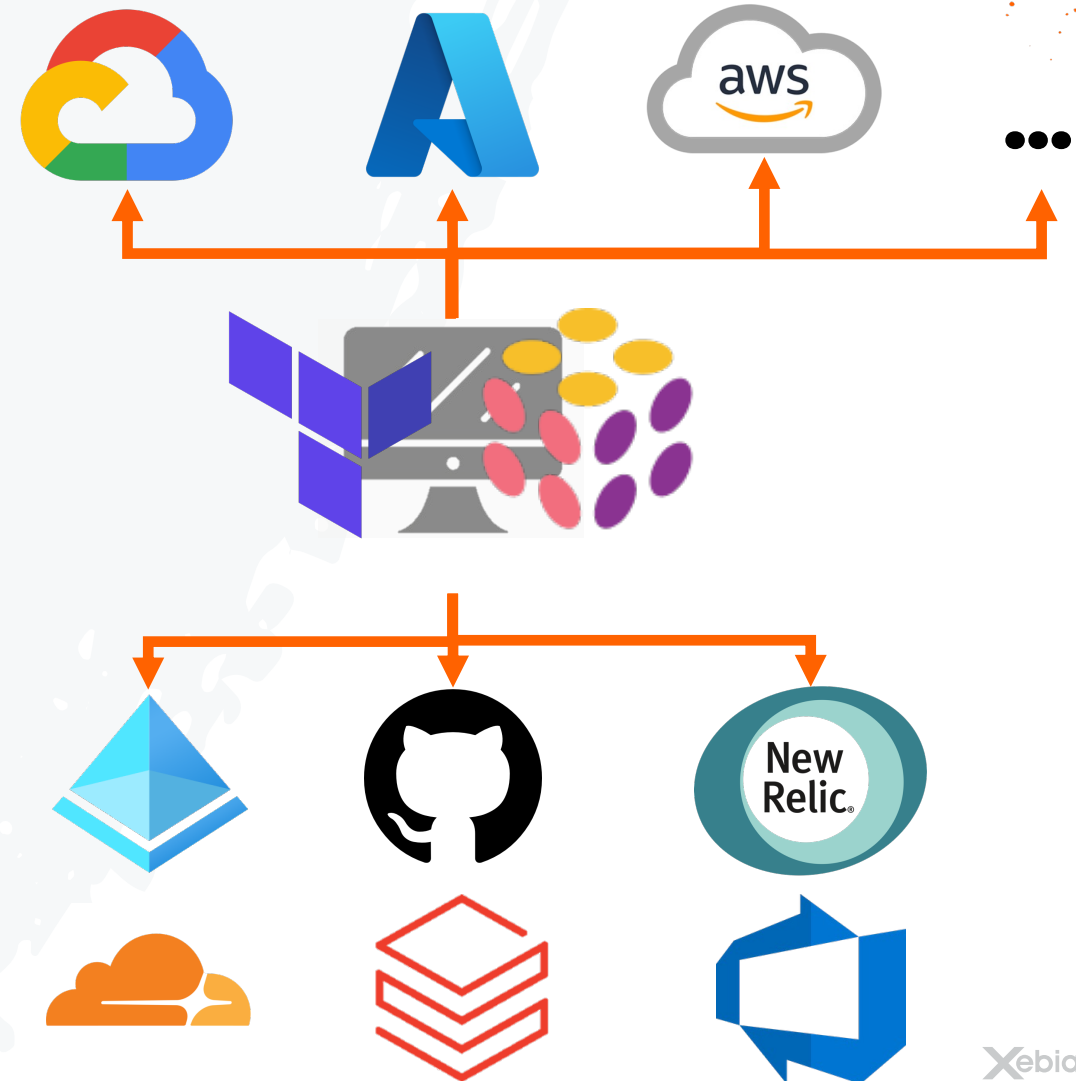
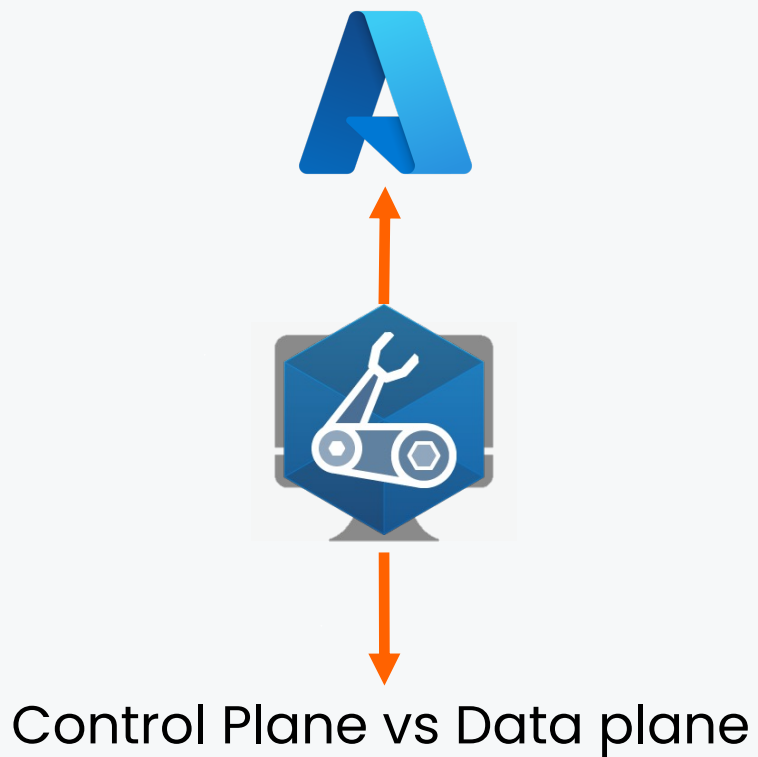
To perform exactly these actions, run the following command to apply:

```
terraform apply "pnds-ddev-104056-20231008001054-tf.plan"
```

Pulumi execution

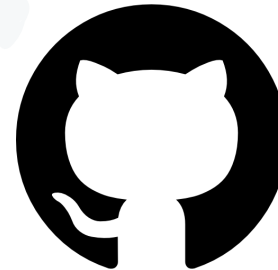


What systems do we want to target?



Terraform and Pulumi Providers

- › Groups resources in logical units per cloud service
- › Official and Community providers
- › Distributed as plugins
- › For Pulumi: Native or Terraform



PagerDuty





Demo!

State

- **Terraform** and **Pulumi** must store state about your managed infrastructure and configuration
- **Bicep** does not have state
- **Terraform**: by default, in a local file named "terraform.tfstate"
 - Can also be stored remotely, which works better in a team environment
- **Pulumi**: by default, in their SaaS offering
- State can make things difficult

[illegible]

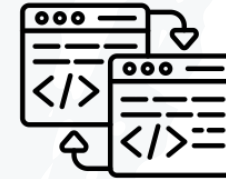
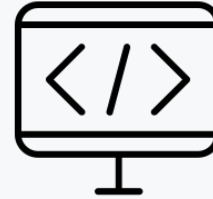
State and secrets

- › State can contain sensitive data: password, private keys, etc.
- › **Terraform** state is stored in plain-text JSON files
- › **Pulumi**: state is encrypted with a per stack key
- › You need to ensure encryption at rest, IAM, TLS connection, IP whitelisting

```
{
  "mode": "managed",
  "type": "azurerm_key_vault_secret",
  "name": "sql_server_admin_password",
  "provider": "provider[\"registry.terraform.io/hashicorp/azurerm\"]",
  "instances": [
    {
      "schema_version": 0,
      "attributes": {
        "content_type": "Managed by Terraform",
        "expiration_date": "2023-03-12T09:13:17Z",
        "id": "https://kv-tfsessionadvanced-dev.vault.azure.net/secrets/sql-admin-password",
        "key_vault_id": "/subscriptions/f0e483fc-9d2f-4a4b-8aee-887a39...",
        "name": "sql-admin-password",
        "resource_id": "/subscriptions/f0e483fc-9d2f-4a4b-8aee-887a39...",
        "value": "GUiYZ9MSYbv6PX4I33FbNsvVtqVx48aI",
        "versionless_id": "https://kv-tfsessionadvanced-dev.vault.azure.net/secrets/sql-admin-password"
      }
    }
  ]
}
```

The power of Pulumi

You can use your favorite programming language: TypeScript, JavaScript, Python, Go, C#, F# Visual Basic, Java, and YAML



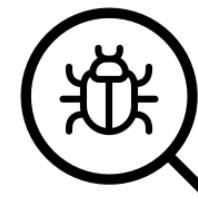
Refactoring

User the power of your favorite IDE



(Unit) testing

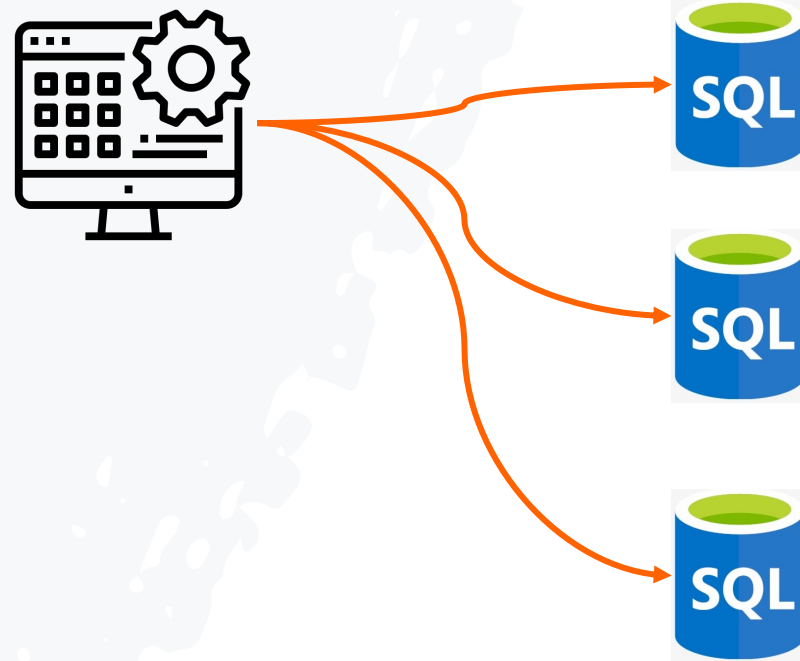
Monitoring / Logging



Debugging

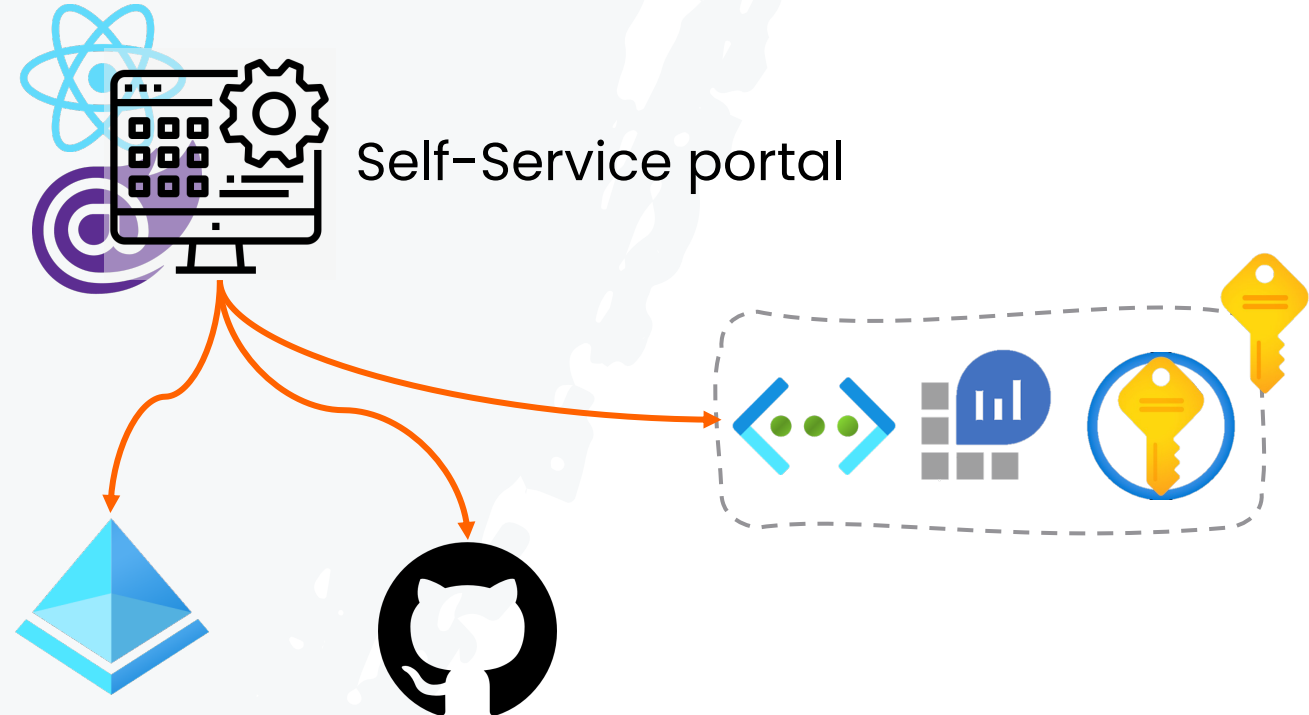
The power of Pulumi

Have applications create
infrastructure

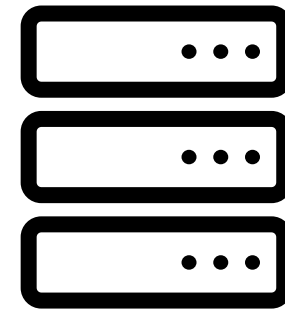
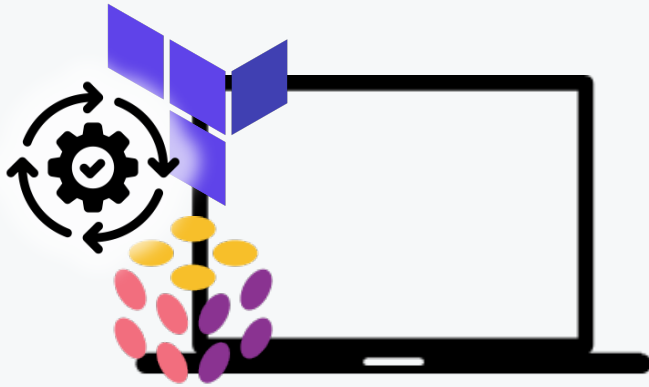
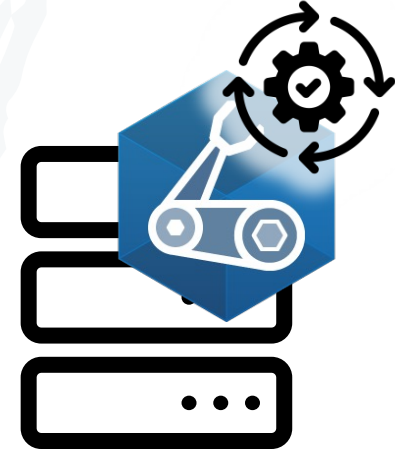


The power of Pulumi

Self-service tooling by
platform teams



Server vs Client-side execution



Ecosystem

- › Azure DevOps / GitHub Actions
- › Checkov
- › Linter with a lot of built-in rules (ARM TTK)
- › PSRule
- › Bicep test framework (experimental)



- › Azure DevOps / GitHub Actions
- › Checkov / TFSec
- › TFLint
- › Terratest
- › Terradocs
- › Git pre-commit
- › Infracost



- › Azure DevOps / GitHub Actions
- › Test Frameworks like NUnit, XUnit
- › SonarCloud



Community – Stackoverflow

terraform

16593 questions 17 asked today, 103 this week

pulumi

422 questions 5 asked this week, 14 this month

azure-bicep

511 questions 7 asked this week, 44 this month

This is not all Azure

This is all Azure

Community – Stackoverflow

🔍 terraform-provider-azur

terraform-provider-azure

1946 questions 11 asked this week, 39 this month

🔍 pulumi-azure

pulumi-azure

36 questions 34 asked this year

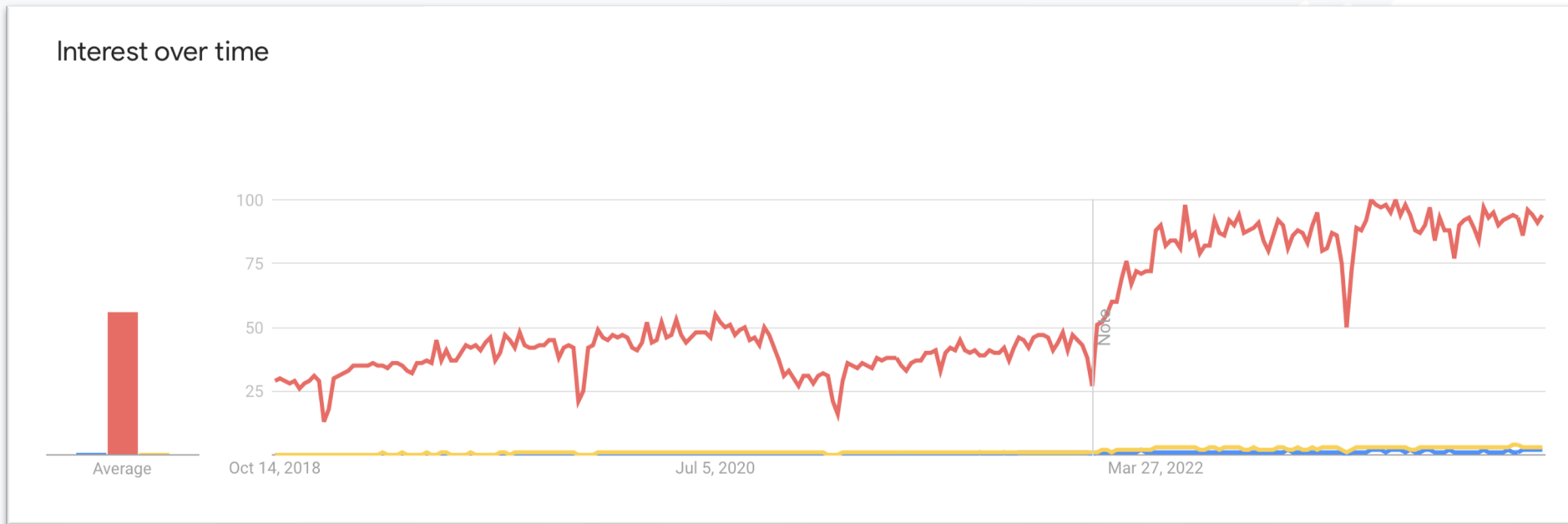
🔍 bicep

azure-bicep

511 questions 7 asked this week, 44 this month

Community – Google Search Trends

Searched for: terraform, azure bicep and Pulumi, last 5 year



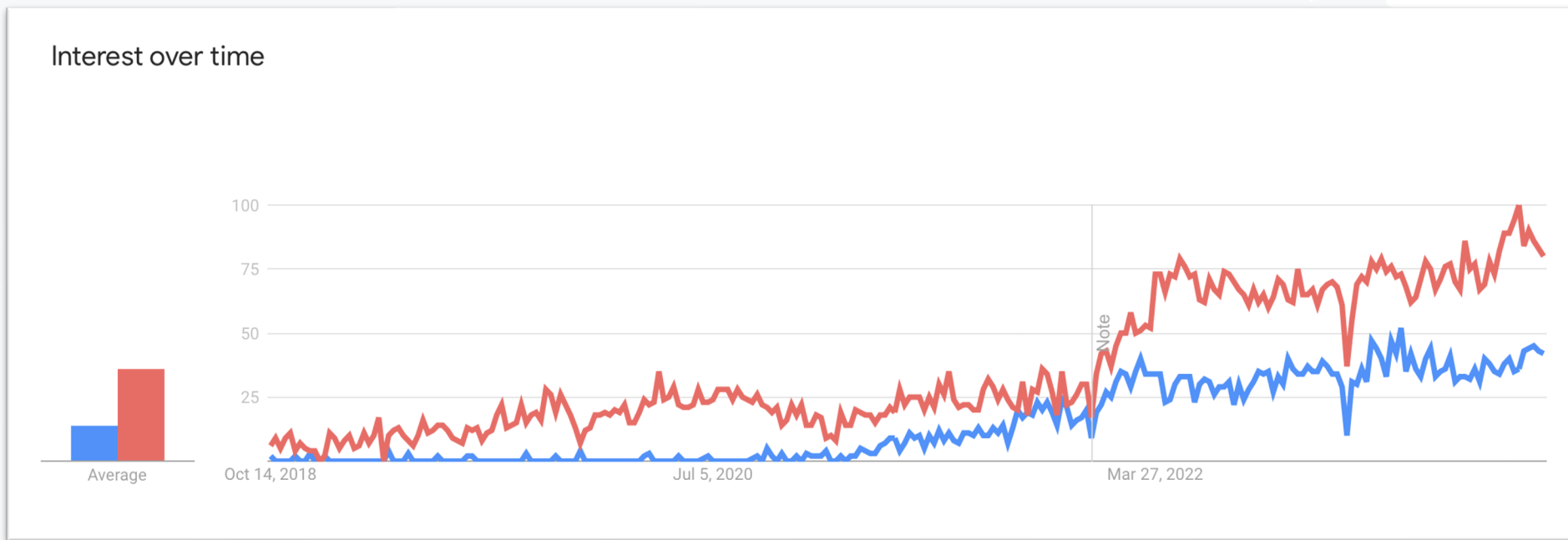
Terraform

Bicep

Pulumi

Community – Google Search Trends

Searched for: azure bicep and Pulumi, last 5 year



Bicep

Pulumi

Do we have a winner?



Simplicity	+++	++	+
Power of DSL / Language	+	++	+++
Multi Cloud	-	++	+++
Developer Experience	+++	+	+++
Deployment	++	+++	+++
Providers	-	++	+++
Community	++	+++	+
Ecosystem	++	+++	+
Build your own workflow	+	+	+++

Thank you!



Erwin Staal

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